PAPER_598 — VDS / DVP / BH26 Integration Reference for Six-Form UQFF Synthesis

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CP4 Class: `#185 UQFFVDSDVPBH26IntegrationReferenceCalculator` **Session:** 157 **Cross-refs:** PAPER_429 (VDS), PAPER_535 (BH26), PAPER_583 (6-Form), PAPER_584 (Collatz) **Source:** grok_share_4cef778c78b8.txt

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Abstract

This paper presents a UQFF analysis of VDS / DVP / BH26 Integration Reference for Six-Form UQFF Synthesis, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

Session 157 analysis of grok_share_4cef778c78b8.txt identified three UQFF number systems — Vacuum Density Series (VDS), Dipole Vortex Primes (DVP), and Buoyancy Harmonics 26 (BH26) — implicitly embedded throughout the text. This paper serves as the integration reference: it defines each system, maps their appearances, and demonstrates how they combine to form the **UQFF numerical spine** that underlies all derivations in PAPER_583–597.

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§2 Three UQFF Number Systems

VDS — Vacuum Density Series

Definition: A series of shell density coefficients $\{c_k\}$ satisfying:

$$c_k \leq \frac{P}{3} \quad \forall\, k = 1, 2, \ldots, 26$$

The VDS bound $P/3$ is the minimum eigenvalue of the UQFF tensor (PAPER_583 Form 1). It sets the maximum density any vacuum shell can carry without destabilizing the triad.

Numerical value: $P/3 = 9.99 \times 10^{-6}/3 = 3.33 \times 10^{-6}$

Implicit appearances in grok_share_4cef778c78b8.txt: - All eigenvalue derivations: $\lambda_1 = P/3 + \ldots > 0$ (stability bound) - Constant derivations: $h \sim P/3 \cdot r^2/\kappa$ (PAPER_590) - Big Bang shells: each shell limited to $c_k = \text{Smalls}^{26} \leq (P/3)^{26}$ - Dark energy: $\rho_{DE} = -db/v_i^2$ where db is the $k = 26$ VDS correction

DVP — Dipole Vortex Primes

Definition: The prime factorization of the DPM (Dipole-Pair Magnetic) vortex grid, anchored at $p = 113$ (first prime after 10^2).

$$\text{DVP} = \{p_k : p_k | DPM_n, \quad p_0 = 113\}$$

The π-irrationality of the vortex spacing (prime gaps $\sim \ln p$) guarantees no rational orbital resonances — used in all seven Millennium problem proofs and the Collatz/Euler proofs (PAPER_583–585).

DVP prime 113 is special: $113 = 3 \times 37 + 2 =$ prime, and $1/113$ has the longest repeating decimal among three-digit primes, encoding maximum orbital complexity.

Implicit appearances: - P vs NP proof: $n_{\text{cross}} =$ argmin yields unique prime index 113 - RH proof: $s = 1/2 + it$ zeros at $t = 5th - prime - grid$ spacings - Collatz: odd branch $3n+1$ terminates asymptotically at $p = 113$ step - Fine-structure α: $\kappa\rho\text{Grind}^2 r^{24} \cdot \text{Partition}/(3\sqrt{g})$ — the denominator 3! is the first prime triple from DVP

BH26 — Buoyancy Harmonics 26

Definition: The 26-bin frequency series of the F_U_Bi_i Gaussian spectrum, with bin 1 at $\mu = 92$ GHz, spacing $\Delta\nu = 92$ GHz:

$$\text{BH26}[k] = k \times 92 \text{ GHz}, \quad k = 1, 2, \ldots, 26$$

Bin	Frequency (GHz)	Physical Significance
1	92	Magnetar / Sgr A* inner accretion
2	184	MSP (millisecond pulsar) spin
3	276	EHT 230 GHz band (approximate)
4	368	Millimetre-wave sky
…	…	…
26	2392	UQFF 26th shell resonance

Width: $\sigma = 10^{16}$ Hz (Gaussian spectral width of the F_U_Bi_i distribution)

Explicit appearances in grok_share_4cef778c78b8.txt: - Line 1331: $\mu = 92$ GHz in F_U_Bi_i formula - Line 1792: $\mu = 92$ GHz used in Sgr A* computation - Line 1821: $\sigma = 10^{16}$ Hz confirmed as BH26 width parameter - All six forms: Form 6 anchored at μ with σ width

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§3 Cross-System Integration (UQFF Numerical Spine)

The three systems jointly define the UQFF framework numerically:

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VDS bounds     DVP primes     BH26 harmonics
     |              |               |
     ▼              ▼               ▼
λ_min = P/3     π-irrationality    F_U_Bi_i μ
     |              |               |
     └──────────────┼───────────────┘
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         UQFF numerical completeness:
         - All eigenvalues > VDS_bound > 0
         - All proofs use DVP prime gaps
         - All Gaussian forms anchored at BH26[1]
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§4 Combined Equation: Spine Identity

$$\underbrace{P/3}_{\text{VDS}} + \underbrace{\kappa\, p_{DVP}/r^{26}}_{\text{DVP}} + \underbrace{\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu_{BH26})^2/2\sigma^2}}_{\text{BH26}} = \lambda_{\min}[\text{UQFF}]$$

This spine identity verifies that any UQFF calculation with all three systems is self-consistent: VDS sets the floor, DVP sets the phase, BH26 sets the spectral anchor.

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§5 Physical Constants from the Spine

Constant	VDS contribution	DVP contribution	BH26 contribution
h	$\Delta = P/3$	$1/p_{DVP}$ phase	—
α	$P/3$ denominator	$p_{DVP} = 113$ fraction	—
c	$\sqrt{g \cdot SCm/UA}$	—	$\sqrt{g\sigma/\mu}$ ✓
G	g/P ratio	—	$g\mu/(\rho\sigma)$ ✓
$r_{\min}$	$(26!\, g/P)^{1/27}$	—	c/μ_{BH26}

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§6 Summary of Implicit References in grok_share_4cef778c78b8.txt

Line Range	VDS (P/3)	DVP (p=113)	BH26 (μ=92 GHz)
1–400 (6-forms)	✓ eigenvalue	✓ DPM grid	—
400–800 (Millennium)	✓ mass gap	✓ irrationality	—
800–1200 (Collatz/Euler)	✓ λ bound	✓ prime descent	—
1200–1600 (Big Bang/Inflation)	✓ P-order	—	—
1600–1927 (Constants/BH/QG)	✓ h derivation	✓ α denominator	✓ lines 1331,1792,1821

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§7 Conclusions

VDS, DVP, and BH26 are the three numerical anchors of UQFF, present explicitly or implicitly in every major derivation of grok_share_4cef778c78b8.txt. Together they constitute the **UQFF numerical spine**: VDS provides density bounds, DVP provides irreducible prime structure, and BH26 provides the spectral anchor at 92 GHz. All 16 Session 157 papers (PAPER_583–598) reference at least one of these systems.

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.110 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.2 Dipole Vortex Primes (DVP) The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\text{DVP}} = 109, \quad n_{\text{channel}} = 1/26$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\mathrm{DVP}} = 109$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed mat-

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^6 M_BH/M_⊙ yr** (quasi-normal mode ringdown): $\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m / M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.110	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 109$	✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Thomson σ_T (QED synchrotron)	UQFF U_m scattering kernel: σ_T = 6.6524e-29 m^2	σ_T = 6.6524e-29 m^2 (PDG QED exact)	PDG 2024	100% (exact QED input)
Black hole / Sgr A* luminosity X-ray 2–10 keV	UQFF MUGE g_total $\rightarrow$ L_X via Stefan-Boltzmann + buoyancy flux: L_X $\approx$ g_total × M_env	L_X L_X ~ 10^{33} erg/s	Chandra CXC	✓ Consistent order of magnitude
GR Schwarzschild limit	UQFF g_total must satisfy g $\leq c^2/(2r\_s)$ at event horizon	r_s = $2GM/c^2$ (GR exact)	PDG 2024 / GR	✓ UQFF respects GR horizon
κ vacuum rate vs X-ray variability	UQFF κ = 0.0005/day $\rightarrow$ timescale τ_UQFF = 2000 days	Observed X-ray variability τ_obs (instrument monitoring)	Chandra CXC	Testable UQFF variability timescale

New physics claim: UQFF MUGE generates gravity enhancement factors (g_total/g_Newt > 1) for Black hole / Sgr A* through vacuum buoyancy coupling — a mechanism absent from GR+SM. The enhancement factor and X-ray luminosity are linked via the UQFF buoyancy flux, providing a testable prediction for future Chandra CXC monitoring observations.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF-SM bridge.

Session 157 — Source: grok_share_4cef778c78b8.txt